

Observation

A Schema.org Type

This term is in the "new" area - implementation feedback and adoption from applications and websites can help improve our definitions.

Thing > **Intangible** > **Observation**

Thing > **Intangible** > **StructuredValue** > **QuantitativeValue** > **Observation**

[more...]

Instances of the class **Observation** are used to specify observations about an entity at a particular time. The principal properties of an **Observation** are **observationAbout**, **measuredProperty**, **statType**, **[[value]]** and **observationDate** and **measuredProperty**. Some but not all Observations represent a **QuantitativeValue**. Quantitative observations can be about a **StatisticalVariable**, which is an abstract specification about which we can make observations that are grounded at a particular location and time.

Observations can also encode a subset of simple RDF-like statements (its **observationAbout**, a **StatisticalVariable**, defining the **measuredProperty**; its **observationAbout** property indicating the entity the statement is about, and **value**)

In the context of a quantitative knowledge graph, typical properties could include **measuredProperty**, **observationAbout**, **observationDate**, **value**, **unitCode**, **unitText**, **measurementMethod**.

Property	Expected Type	Description
Properties from Observation		
marginOfError	QuantitativeValue	A marginOfError for an Observation .
measuredProperty	Property	The measuredProperty of an Observation , typically via its StatisticalVariable . There are various kinds of applicable Property : a schema.org property, a property from other RDF-compatible systems, e.g. W3C RDF Data Cube, Data Commons, Wikidata, or schema.org extensions such as GS1's .
measurementDenominator	StatisticalVariable	Identifies the denominator variable when an observation represents a ratio or percentage.
measurementMethod	DefinedTerm or MeasurementMethodEnum or Text or URL	A subproperty of measurementTechnique that can be used for specifying specific methods, in particular via MeasurementMethodEnum .
measurementQualifier	Enumeration	Provides additional qualification to an observation. For example, a GDP observation measures the Nominal value.
measurementTechnique	DefinedTerm or MeasurementMethodEnum or Text or URL	A technique, method or technology used in an Observation , StatisticalVariable or Dataset (or DataDownload , DataCatalog), corresponding to the method used for measuring the corresponding variable(s) (for datasets, described using variableMeasured ; for Observation , a StatisticalVariable). Often but not necessarily each variableMeasured will have an explicit representation as (or mapping to) an property such as those defined in Schema.org, or other RDF vocabularies and "knowledge graphs". In that case the

Property	Expected Type	Description
		subproperty of variableMeasured called measuredProperty is applicable. The measurementTechnique property helps when extra clarification is needed about how a measuredProperty was measured. This is oriented towards scientific and scholarly dataset publication but may have broader applicability; it is not intended as a full representation of measurement, but can often serve as a high level summary for dataset discovery. For example, if variableMeasured is: molecule concentration, measurementTechnique could be: "mass spectrometry" or "nmr spectroscopy" or "colorimetry" or "immunofluorescence". If the variableMeasured is "depression rating", the measurementTechnique could be "Zung Scale" or "HAM-D" or "Beck Depression Inventory". If there are several variableMeasured properties recorded for some given data object, use a PropertyValue for each variableMeasured and attach the corresponding measurementTechnique. The value can also be from an enumeration, organized as a MeasurementMethodEnumeration.
observationAbout	Place or Thing	The observationAbout property identifies an entity, often a Place , associated with an Observation .
observationDate	DateTime	The observationDate of an Observation .
observationPeriod	Text	The length of time an Observation took place over. The format follows <code>P[0-9]*[Y M D h m s]</code> . For example, P1Y is Period 1 Year,

Property	Expected Type	Description
		P3M is Period 3 Months, P3h is Period 3 hours.
variableMeasured	Property or PropertyValue or StatisticalVariable or Text	The variableMeasured property can indicate (repeated as necessary) the variables that are measured in some dataset, either described as text or as pairs of identifier and description using PropertyValue, or more explicitly as a StatisticalVariable .
Properties from QuantitativeValue		
	PropertyValue	A property-value pair representing an additional characteristic of the entity, e.g. a product feature or another characteristic for which there is no matching property in schema.org.
additionalProperty		Note: Publishers should be aware that applications designed to use specific schema.org properties (e.g. https://schema.org/width , https://schema.org/color , https://schema.org/gtin13 , ...) will typically expect such data to be provided using those properties, rather than using the generic property/value mechanism.
maxValue	Number	The upper value of some characteristic or property.
minValue	Number	The lower value of some characteristic or property.
unitCode	Text or URL	The unit of measurement given using the UN/CEFACT Common Code (3 characters) or a URL. Other codes than the UN/CEFACT Common Code may be used with a prefix followed by a colon.
unitText	Text	A string or text indicating the unit of measurement. Useful if you cannot provide a standard unit code for unitCode .

Property	Expected Type	Description
value	Boolean or Number or StructuredValue or Text	The value of a QuantitativeValue (including Observation) or property value node. - For QuantitativeValue and MonetaryAmount, the recommended type for values is 'Number'. - For PropertyValue, it can be 'Text', 'Number', 'Boolean', or 'StructuredValue'. - Use values from 0123456789 (Unicode 'DIGIT ZERO' (U+0030) to 'DIGIT NINE' (U+0039)) rather than superficially similar Unicode symbols. - Use '.' (Unicode 'FULL STOP' (U+002E)) rather than ',' to indicate a decimal point. Avoid using these symbols as a readability separator.
valueReference	DefinedTerm or Enumeration or MeasurementTypeEnumeration or PropertyValue or QualitativeValue or QuantitativeValue or StructuredValue or Text	A secondary value that provides additional information on the original value, e.g. a reference temperature or a type of measurement.
Properties from Thing		
additionalType	Text or URL	An additional type for the item, typically used for adding more specific types from external vocabularies in microdata syntax. This

Property	Expected Type	Description
		is a relationship between something and a class that the thing is in. Typically the value is a URI-identified RDF class, and in this case corresponds to the use of <code>rdf:type</code> in RDF. Text values can be used sparingly, for cases where useful information can be added without their being an appropriate schema to reference. In the case of text values, the class label should follow the schema.org style guide .
alternateName	Text	An alias for the item.
description	Text or TextObject	A description of the item.
disambiguatingDescription	Text	A sub property of description. A short description of the item used to disambiguate from other, similar items. Information from other properties (in particular, name) may be necessary for the description to be useful for disambiguation.
identifier	PropertyValue or Text or URL	The identifier property represents any kind of identifier for any kind of Thing , such as ISBNs, GTIN codes, UUIDs etc. Schema.org provides dedicated properties for representing many of these, either as textual strings or as URL (URI) links. See background notes for more details.
image	ImageObject or URL	An image of the item. This can be a URL or a fully described ImageObject .
mainEntityOfPage	CreativeWork or URL	Indicates a page (or other CreativeWork) for which this thing is the main entity being described. See background notes for details. Inverse property: mainEntity

Property	Expected Type	Description
name	Text	The name of the item.
potentialAction	Action	Indicates a potential Action, which describes an idealized action in which this thing would play an 'object' role.
sameAs	URL	URL of a reference Web page that unambiguously indicates the item's identity. E.g. the URL of the item's Wikipedia page, Wikidata entry, or official website.
subjectOf	CreativeWork or Event	A CreativeWork or Event about this Thing. Inverse property: about
url	URL	URL of the item.

Instances of **Observation** may appear as a value for the following properties

Property	On Types	Description
diseaseSpreadStatistics	SpecialAnnouncement	Statistical information about the spread of a disease, either as WebContent , or described directly as a Dataset , or the specific Observations in the dataset. When a WebContent URL is provided, the page indicated might also contain more such markup.

Source

<https://github.com/schemaorg/schemaorg/issues/2291>

Examples

📄 Example 1

No Markup

JSON-LD

Structure

Example encoded as JSON-LD in a HTML script tag.

```
<script type="application/ld+json">
[
  {
    "@context": "https://schema.org/",
    "@type": "StatisticalVariable",
    "@id": "Median_Height_Person_Female",
    "name": "Median height of women",
    "populationType": {"@id": "Person"},
    "measuredProperty": {"@id": "height"},
    "statType": {"@id": "median"},
    "gender": {"@id": "Female"},
    "numConstraints": 1,
    "constrainingProperty": {"@id": "gender"}
  },
  {
    "@context": "https://schema.org/",
    "@id": "Observation_Median_Age_Person_Female_SanAntonio_TX_2014",
    "@type": "Observation",
    "name": "Median height of women in San Antonio, Texas in 2014",
    "description": "An Observation of the StatisticalVariable Median_Height_Person_Female in location: San Ar
```



```
"variableMeasured": { "@id": "Median_Height_Person_Female" },  
"observationAbout": { "@id": "https://www.wikidata.org/entity/Q975" },  
"observationDate": 2014,  
"value": 160,  
"unitCode": "CMT"  
}  
]  
</script>
```